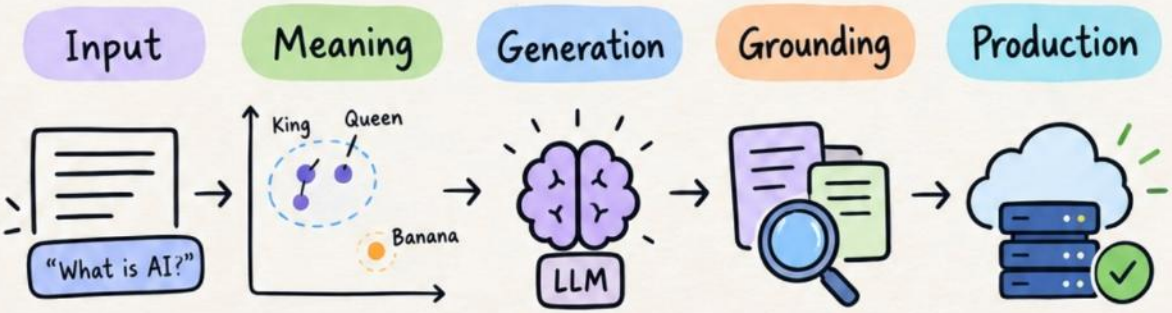
10 AI CONCEPTS YOU MUST KNOW BEFORE YOUR INTERVIEW

From prompt to production, in 10 swipes.

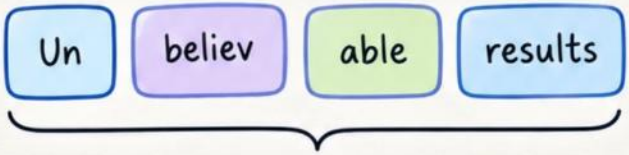


Swipe →

Stage 1 INPUT

The model starts with text.

1 TOKENS



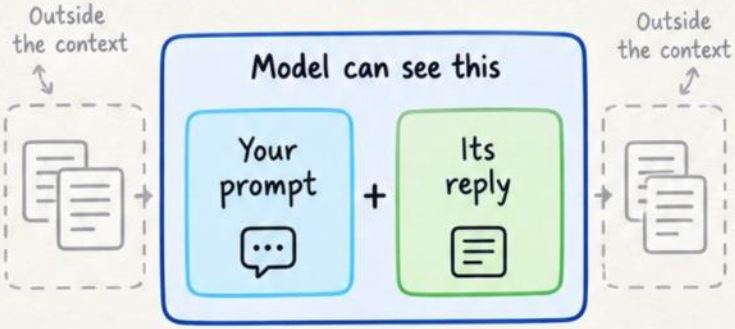
4 tokens ≠ 2 words

(note: illustrative, tokenizers vary)

Before processing, text is split into tokens: words, word pieces, or punctuation.

Limits and pricing are counted in tokens, not words.

2 CONTEXT WINDOW

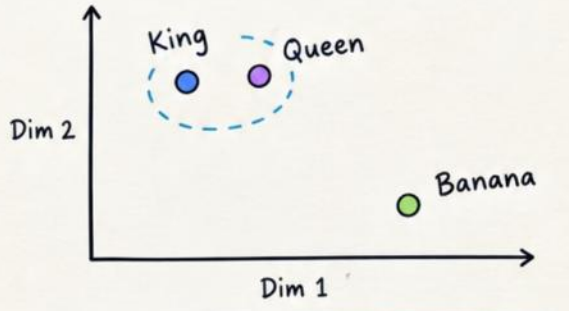


The model can only use what fits in one request: your prompt plus its reply.

Anything outside the window is invisible to it.

Stage 2 MEANING

3 EMBEDDINGS 1010

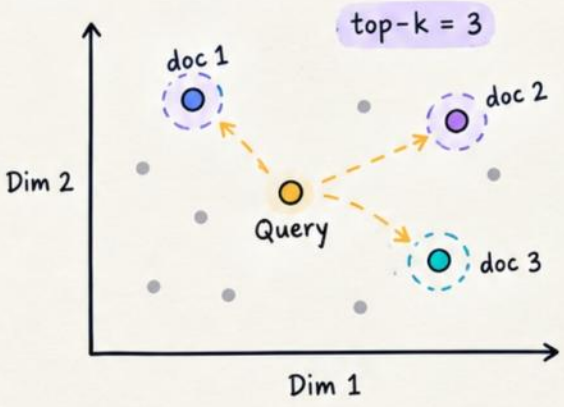


Closer ≈ more similar meaning

An embedding model turns text into a list of numbers.

Texts with similar meaning end up with vectors that sit close together.

4 VECTOR SEARCH

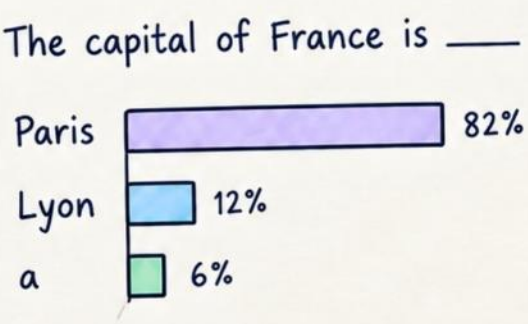


Your query is embedded too, then compared against stored vectors.

The closest matches come back, ranked by similarity.

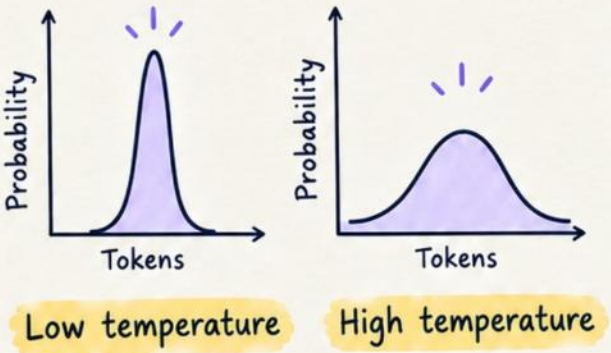
Stage 3 GENERATION

5 NEXT-TOKEN PREDICTION



An LLM doesn't write a whole answer at once. It scores every possible next token, picks one, and repeats.

6 SAMPLING



Temperature sets how strongly the model favors its top choice.

Top-p limits picks to the smallest set of tokens whose probabilities add up to p.

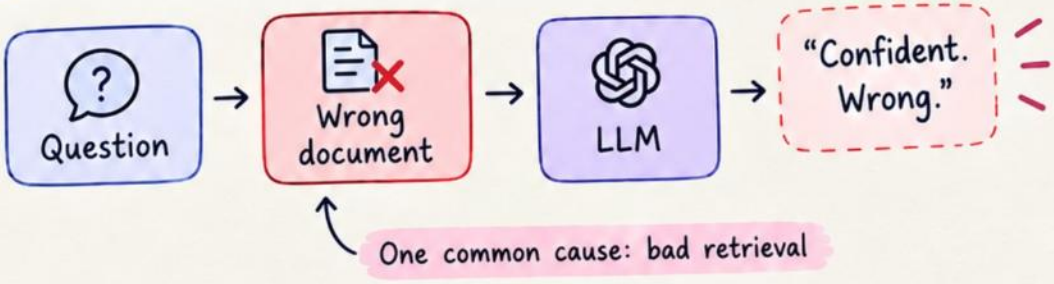
Stage 4 GROUNDING

7 RAG



Instead of relying on memory, the system retrieves relevant documents first and hands them to the model along with your question.

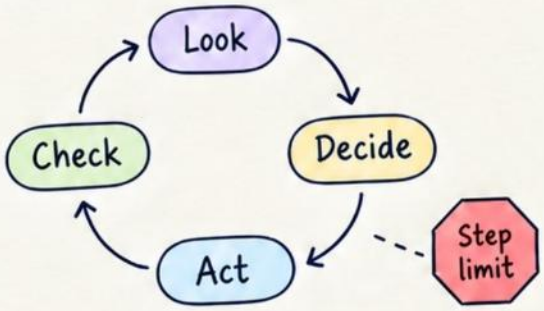
8 HALLUCINATION



The model predicts plausible text, not verified truth. When evidence is missing or wrong, it can still answer with full confidence.

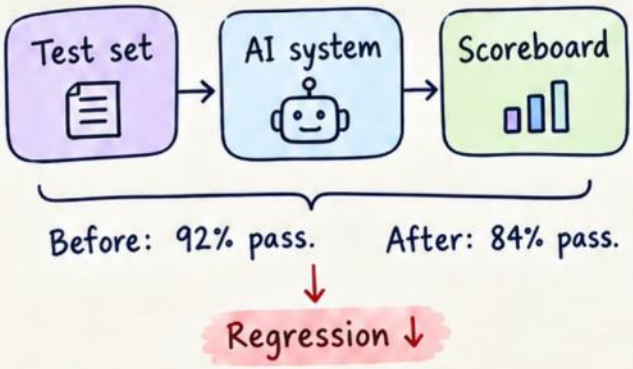
Stage 5 PRODUCTION

9 AI AGENTS



An agent runs the model in a loop: decide the next action, call a tool, read the result, repeat until done or a limit is hit.

10 EVALS



You run a fixed set of test cases against your system and score the results, so every change shows up as better or worse.

Learn the knobs. Learn what breaks.
Save this. Comment CONCEPTS for the one-page cheat sheet.